Pranaya Pratik Das

Physics, Research Scholar

December 22, 1992

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Online Platforms

Pranaya Pratik Das

Pranaya-Das

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Skills

Programming Software

- Mathematica
- FOTRAN
- Matlab
- Python
- Julia

Text editing Software

- MS Office
- TexStudio
- LibreOffice

Operating Systems

- Microsoft Windows
- Ubuntu
- MacOS

Education

Academic Curriculum

- 10^{th}. *June 2008* — BSE, Odisha

  Achievement: 100 % *in Mathematics*

- $+2$. **Science** *June 2010* — CHSE, Odisha
- **B.Sc. in Physics** *July 2010 – June 2013* — BJB Auto. College, Utkal University
- **M.Sc. in Physics** *August 2014 – August 2016* — CBSH, OUAT
- **Ph.D. in Physics** *July 2019–2025* — NIT Rourkela

  Thesis Topic: *Diagnosis of quantum chaos in perturbed quantum wells and billiards*

Awards and Achievements

Scholarships:

- P.G. Meritorious Scholarship (2014-16), Institute of Mathematics and Applications (IMA).
- Medhabruti Scholarship (2014)

Certificates:

- IAPT(2012)
- Spring College in the Physics of Complex Systems. *Awarded by ICTP.*
- Secrets of getting published in high impact factor journals. *Awarded by Wiley.*

Qualified Entrances:

- TIFR (2015-16)
- GATE (2019-21)

Research & Publications

Recent Publications

2025: **Pranaya Pratik Das and Biplab Ganguli**. *"Harmonic oscillator under nonlinear spatial perturbations: A study via Out-of-Time-Order Correlators"*. **(On going)**

2025: **Vinesh Vijayan, Pranaya Pratik Das, K Hariprasad and P Sathish Kumar**. *"Cyclically Symmetric Thomas Oscillators As Swarmalators: A model for Active Fluids and Pattern Formation."*. **Communications in Nonlinear Science and Numerical Simulation Volume 152, Part B, January 2026, 109216**
DOI: https://doi.org/10.1016/j.cnsns.2025.109216

2025: **Pranaya Pratik Das and Biplab Ganguli**. *"Out-of-Time-Order Correlation in perturbed quantum wells"*. **Eur. Phys. J. D (2025) 79 :74**
DOI: https://doi.org/10.1140/epjd/s10053-025-01025-7

2025: **Pranaya Pratik Das, Tanmayee Patra, and Biplab Ganguli**. *"Manifestations of chaos in billiards: the role of mixed curvature"*.
DOI: https://doi.org/10.48550/arXiv.2501.08839

2024: **Bhaskar Shukla, Pranaya Pratik Das, David Dudal, and Subhash Mahapatra**. *"Interplay between the Lyapunov exponents and phase transitions of charged AdS black holes."*. **Phys. Rev. D 110, 024068**
DOI: 10.1103/PhysRevD.110.024068

2022: **Vinesh Vijayan, & and Pranaya Pratik Das**. *"Dynamics of a charged Thomas oscillator in an external magnetic field"*. **Physica Scripta, 97(11), 115207**.
DOI 10.1088/1402-4896/ac99ab

Research Interest

- Classical Chaos
- Quantum Chaos
- Quantum Scars
- Quantum Speed Limits
- Billiards Dynamics
- Black Hole Phase Transition

Please turn over...

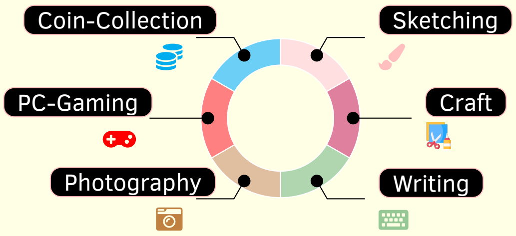
Pranaya Pratik Das

Physics, Research Scholar

About Me

I am a research scholar specialising in quantum chaos and nonlinear dynamics, focusing particularly on out-of-time-order correlation (OTOC), spectral complexity, scrambling diagnostics, and quantum scars. By collaborating on projects related to black hole phase transitions, chaos near black holes, and the swarming dynamics of active particles, my work bridges multiple domains.

Hobbies



Languages

- Odia
- Hindi
- English

References

[1] **Prof. Biplab Ganguli**
Department of Physics and Astronomy, NIT Rourkela
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biplab62g@gmail.com

[2] **Prof. Subhash C. Mahapatra**
Department of Physics and Astronomy, NIT Rourkela
mahapatrasub@nitrkl.ac.in
subhashmahapatra@gmail.com

[3] **Prof. Mithun Biswas**
Department of Physics and Astronomy, NIT Rourkela
biswasm@nitrkl.ac.in

Teaching Assistance

Laboratory

- **TA in Thermal Physics Lab** *2022–2023, Spring Semester* — Prof. B. Ganguli
  - Conducted various classes, experiments, evaluated copies and and taken vivas.
- **TA in B.Tech Physics Lab** *2022–2023, Spring Semester* — Prof. B. Ganguli
  - Conducted various classes, experiments, evaluated copies and and taken vivas.
- **TA in B.Tech Physics Lab** *2022–2023, Autumn Semester* — Prof. I. Banarjee
  - Conducted various classes, experiments, evaluated copies and and taken vivas.
- **TA in B.Tech Physics Lab (Online)** *2021–2022, Spring Semester* — Prof. S. K. Bisoi
  - Conducted various online classes, seminars and evaluated copies.
- **TA in B.Tech Physics Lab (Online)** *2021–2022, Autumn Semester* — Prof. P. Mahanandia
  - Conducted various online classes and evaluated copies.

Theoretical

- **Co-supervised a M.Sc. project** *2023–2024* — Prof. B. Ganguli
  - Successfully co-supervised an M.Sc. project for Zubair Ahmad Kumar (422PH2069).
  - Successfully co-supervised an M.Sc. project for Vivek Sheoran (422PH2082).
- **Co-supervised a M.Sc. project** *2022–2023* — Prof. B. Ganguli
  - Successfully co-supervised an M.Sc. project for Karishma Kujur (421PH2125).
  - Successfully co-supervised an M.Sc. project for Ayush Sahu (418PH5033).

Conferences and Schools

- **60 Years of DFT: Advancements in Theory & Computation** *2024* — IIT Mandi, India
  - Poster Presentation
- **HPC Symposium** *2024* — NIT Rourkela, India
  - Poster Presentation
- **Integrability, Deformations and Chaos** *2023* — OIST, Onna, Okinawa, Japan
  - Attended online
- **School on Quantum Chaos** *2023* — ICTP-SAIFR, São Paulo, Brazil
  - Attended online
- **Complex Lagrangian Problems of Particles in Flows** *2022* — ICTS-TIFT, India
  - Attended online
- **Spring College in the Physics of Complex Systems** *2022* — ICTP, Trieste, Italy
  - Attended online